

A Parsimonious Bayesian Framework with Closed-Form Logit Posterior for Brazilian Presidential Runoff Forecasting

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Abstract

We develop and validate a Bayesian forecasting framework for the second round of Brazilian presidential elections, yielding a single probability $\theta = \Pr(A \text{ wins runoff})$ via a closed-form Gaussian posterior in logit space. The framework fuses a *parsimonious two-source core* (opinion polls and government approval) selected empirically after drop-one source-sensitivity on six historical runoffs. The model was substantially revised in v6.3 after an internal peer review uncovered (i) data leakage in the v5/v6 house-effects corrections, (ii) precision double-counting in the poll-likelihood fusion, and (iii) a poorly-identified treatment of candidates whose party recently held — but does not currently hold — the presidency. Institute-specific corrections were **abandoned** after empirical demonstration that even an honest leave-one-out pipeline degrades the backtest versus raw polls. Walk-forward backtesting on six post-2002 runoffs identifies the winner in 6/6 cases with Brier 0.097 and mean absolute share error 2.7 pp. Applied to the 2026 race under candidate uncertainty across four right-wing succession scenarios — Tarcísio de Freitas (PL/REP), Flávio Bolsonaro, Michelle Bolsonaro, and Renan Santos (MBL) — the marginalized point estimate is $\Pr(\text{right wins}) = 0.502$ (Lula 0.498), at the knife’s edge, with a ± 20 pp approval shock moving the marginal by ± 1.3 pp.

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1 A Parsimonious Bayesian Framework with Closed-Form Logit Posterior for Brazilian Presidential Runoff Forecasting

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Changes in v6.3 (May 2026). Following an internal peer review documented in PEER_REVIEW.md, four bugs were identified and corrected, and the institute-bias correction was abandoned:

R7.1 — Data leakage in house-effects corrections (S1 of review). The v5/v6 institute-bias model estimated b_h from residuals $r_{j,h,t} = y_{h,t,\text{real}} - p_{j,h,t}$ over *all* elections $t \in \{2002, \dots, 2022\}$ and then applied those corrected polls back into the *same* elections during walk-forward backtesting, producing an in-sample leak. A leave-one-out reimplementaion (excluding the target election from b_h training) was developed and tested. Empirically, *both* the leaky in-sample correction (Brier 0.132) and the honest LOO correction (Brier 0.115) degrade the backtest relative to raw polls (Brier 0.097 with the v6.3 fixes below). Diagnosis: the model $b_h \sim \text{institute}$ is

mis-specified because the identity of candidate A rotates between PT, PSDB, and PL across elections; aggregating residuals across these introduces systematic mis-correction. **Institute-bias corrections are abandoned in v6.3**; polls are used raw.

R7.2 — Precision double-counting in poll fusion (S2 of review). The flat-fit fusion in `model_2T.fit_2T` computed effective precision as $w_i \cdot k_i / \sigma_i^2$ with $k_i = N_{\text{polls}}$ for opinion polls, but σ_i was already the *pooled* standard deviation $\sigma_{\text{pool}} = (\sum_j 1/\sigma_j^2)^{-1/2}$, which itself incorporates N . The product over-weighted polls by a factor of N , so 252 polls in 2022 dominated approval by a ratio of $\approx 27,000 : 1$. Sensitivity tests of approval ± 20 pp on 2022 produced *exactly* zero variation in $\Pr(A)$. v6.3 replaces the formula with w_i / σ_i^2 (correct for a Gaussian observation already pooled). A precision cap is additionally imposed via $N_{\text{eff,max}} = 30$ “equivalent” polls so that 252 polls never overwhelm the approval signal entirely. The hierarchical path (`fit_2T_hierarchical`, v6.2) did not have this bug.

R7.3 — Incumbency strength as a continuous covariate (M5 of review). The Boolean `incumbent_is_A` flag was replaced with a continuous `incumbent_strength_A` $[0,1]$ that modulates the sign of the approval coefficient: $\text{sign} = 2 \cdot \text{strength} - 1$. In the historical sample this is 0.0 for the canonical incumbent-versus-challenger races (2002 Serra/PSDB, 2006/2010 Lula-Dilma/PT, 2014 Dilma/PT, 2022 Bolsonaro/PL) and 0.5 for 2018 (Haddad/PT inherits a symbolic incumbency after the 2016 PT impeachment, with $\text{sign} = 0$ neutralizing the approval signal — Temer/MDB was the formal incumbent but did not reach the runoff). This eliminates a sign-misapplication that had been masking 2018’s true signal load.

R7.4 — Other (S4 / M1 of review). The transfer-matrix intercept θ_0 in `transfer_matrix` had been silently zeroed via a stray `theta0 * 0` (fixed); the irreducible-variance scaling in the deprecated multiclass 1T path was applying the time scaling to *all* three components rather than to drift alone (fixed and the entire 1T multiclass path was archived to `_deprecated/`).

Updated headline metrics (6 runoffs, 2002–2022): Brier 0.097 (v6.2 had reported 0.132 but the reproducible run with the v6.2 code gives 0.132 — confirming the value with the bug), accuracy 6/6 (genuinely confident, no $P=0.500$ ties), mean absolute share error 2.7 pp. Drop-one shows both core sources are now load-bearing: $\Delta \text{Brier}_{\text{polls}} = +0.016$ and $\Delta \text{Brier}_{\text{appr}} = +0.035$ (the v6.2 model had a *negative* polls delta — an artifact of the leaky corrections). Stress on ± 20 pp Lula approval in 2022 now produces a ± 2.6 pp swing in $\Pr(A)$ (zero in v6.2).

The 2026 forecast is **unchanged** at $\Pr(\text{right}) = 0.502$ because the hierarchical path used for projection did not carry the precision double-count.

Changes from v5.0 to v6.2 (preserved for completeness). Four methodological revisions were introduced:

R1 (parsimonious core). The model is simplified from a six-source fusion to

a **two-source core** (opinion polls and government approval) after head-to-head empirical evidence showed that markets, donations, roll-call cohesion, and demographics contributed ≤ 0.001 to the Brier score and injected ECE-degrading noise in data-sparse elections. The four demoted sources are still computed and reported as **auxiliary out-of-posterior diagnostics**.

R2 (two-way random-effects house effects). The institute-only house-effect model is generalized to a two-way decomposition separating persistent institute bias b_h from election-wide swing δ_t . Empirical decomposition attributes 33.8% of the historical poll-residual variance to election-wide δ_t — a third of what v5 erroneously assigned to institutes. The realigned institute biases cluster much closer to zero, while Veritá’s right-leaning posture and the documented 2014/2018 shy-voter effects emerge cleanly.

R3 (recalibrated source weights — v6.1). A formal sensitivity audit of the historical posterior reveals that in poll-rich elections the approval signal — though structurally large — was *operationally swamped* by poll precision (ratios of 100:1 to 100,000:1 in 2010/2018/2022). Recalibrating $(\bar{\gamma}_{\text{polls}}, \bar{\gamma}_{\text{appr}})$ from (0.55, 0.45) to (0.30, 0.70) recovers the 2010 election (Lula approval +55 pp peak previously masked by noisy polls) and restores 6/6 accuracy with Brier 0.132 (vs. 0.135 in v6.0).

R4 (hierarchical candidate model — v6.2). To resolve the “candidate-blind” pathology in lean v6.1 — under which candidates without dedicated polls collapse to the same probability — we introduce a bivariate Gaussian decomposition $\varphi_k = \varphi_{\text{base}} + \delta_k$. Approval observes φ_{base} ; polls and structural expert anchors observe $\varphi_{\text{base}} + \delta_k$ jointly. The closed-form posterior is multivariate Gaussian, with marginal φ_k recovered analytically. This enables candidate-specific probabilities even when polls are absent, via expert anchors with explicit uncertainty $\sigma_{x,k}$.

Net effect on the 2026 forecast. With the four-candidate succession field (Tarcísio added as the post-Banco Master PL natural successor), a -1.5 pp event-shift on Flávio polls, and a composite empirical anchor for Tarcísio ($0.57 \cdot \text{pop}_{\text{SP}} + 0.51 \cdot \text{pop}_{\text{SP}} = 0.523$ national), the v6.2 marginalized estimate is $\text{Pr}(\text{right wins}) = 0.502$ (Lula 0.498) — a sharp narrowing from v6.1’s 0.541, driven by the now-distinguishable, structurally weak Renan scenario and the empirically anchored Michelle scenario.

1.1 Disclosure regarding the use of artificial intelligence

This working paper was prepared with the assistance of generative artificial-intelligence tools (Anthropic Claude, model family Claude 4.x). The AI assisted in the following tasks: drafting of expository text, generation and refactoring of data-processing and modeling code (Python), running of the validation and diagnostic suites, suggestion of bibliographic references, and stylistic editing of the manuscript.

The methodological choices, modeling decisions, data interpretation, and conclusions are the sole intellectual responsibility of the author. The author has reviewed all AI-generated content, in-

cluding all mathematical derivations, code, and numerical results, and assumes full responsibility for the accuracy and integrity of the paper.

This disclosure follows the recommended practice of major scholarly journals as of 2024–2026 (cf. Nature 2023; Science 2024; JASA editorial guidance) for transparent declaration of generative-AI assistance in the preparation of academic manuscripts.

1.2 Abstract

We develop and validate a Bayesian forecasting framework for the second round of Brazilian presidential elections. The framework yields a single probability,

$$\theta = \Pr(\text{candidate } A \text{ wins runoff}),$$

computed via a **closed-form Gaussian posterior in logit space**. The posterior is built from a *parsimonious two-source core* — opinion polls and government approval — selected empirically after a drop-one source-sensitivity analysis on six historical runoffs showed that four additional sources (prediction markets, donations, roll-call cohesion, demographics) contributed at most 0.001 to the Brier score and degraded the expected calibration error in elections without poll data. Those four sources remain computed and reported as **out-of-posterior diagnostics**. The model is binary by construction, sidestepping the high-dimensional simplex of multi-candidate first-round forecasting. We estimate institute-specific (*house effect*) biases via partial pooling with empirical-Bayes shrinkage (Jackman 2005; Linzer 2013) using 356 historical runoff polls from five elections (2006–2022). The v6.2 specification extends the framework to a **bivariate Gaussian hierarchical decomposition** $\varphi_k = \varphi_{\text{base}} + \delta_k$, separating the macro mood — driven by incumbent approval — from candidate-specific deviation, enabling principled handling of scenarios without dedicated polls via structural expert anchors. Walk-forward backtesting on six post-2002 runoffs shows the v6.1 (flat) specification correctly identifies the winner in 6/6 cases with Brier 0.132, MAE in share 4.5 pp; the v6.2 hierarchical extension preserves 6/6 with a calibration penalty (Brier 0.145) deliberately accepted in exchange for candidate identifiability. We then apply the framework to the 2026 race under candidate uncertainty, marginalizing over four right-wing succession scenarios (Tarcísio de Freitas, Flávio Bolsonaro with a −1.5 pp Banco Master event-shift, Michelle Bolsonaro, Renan Santos). The point estimate is

$$\Pr(\text{right wins 2026 runoff}) = 0.502, \quad \Pr(\text{Lula}) = 0.498,$$

at the knife’s edge — a sharp narrowing from v6.1’s 0.541, driven by the now-distinguishable structurally weak Renan scenario and the empirically anchored Tarcísio composite.

Keywords: election forecasting, Bayesian inference, Brazil, runoff elections, partial pooling, house effects

1.3 Notation

Symbol	Meaning
t	election index (year)
A, B	runoff candidates: A = challenger, B = incumbent / governist
$\theta_t \in [0, 1]$	probability that A wins the runoff
$\varphi_t \in \mathbb{R}$	logit transform: $\varphi_t = \text{logit}(\theta_t) = \log(\theta_t/(1 - \theta_t))$
μ_0, σ_0^2	mean and variance of the structural Gaussian prior on φ_t
ℓ_i	likelihood of source $i \in \{1, \dots, 6\}$
$\hat{\varphi}_i, \sigma_i^2$	point estimate and variance contributed by source i
k_i	effective sample size of source i
$\gamma_i \in [0, 1]$	dynamic source weight (decomposed as $\alpha_i \cdot \beta_i \cdot \bar{\gamma}_i$)
$\mu_{\text{post}}, \sigma_{\text{post}}^2$	posterior mean and variance of φ_t
σ_{irr}^2	irreducible variance (drift + residual house + structural surprise)
τ	forecast horizon in days
$\Phi(\cdot)$	standard normal CDF
$\mathcal{N}(\mu, \sigma^2)$	normal distribution
b_h	house effect (institute-level systematic bias) of pollster h
τ^2	between-institute variance of house effects (hyperparameter)
$r_{j,h,t}$	residual of poll j from pollster h in election t : $s_{A,t}^{\text{real}} - \hat{p}_{A,j,h,t}$

1.4 1. Introduction

Brazil’s two-round presidential election system has decided five of the last six elections in the runoff. This stylized fact motivates the modeling choice that anchors this paper: rather than forecasting multi-candidate first-round vote shares on the $(K - 1)$ -dimensional simplex—which adds complexity without materially improving the question of *who wins the presidency*—we forecast directly the **binary runoff outcome**.

The contribution is methodological: a closed-form Gaussian posterior in logit space that unifies six information sources with explicit weights γ_i , removing the need for MCMC and allowing transparent decomposition of evidence per source. In v6.2 the model is extended hierarchically to permit candidate-specific forecasts even without dedicated polls, via a bivariate Gaussian decomposition into macro mood and candidate deviation. The framework is validated walk-forward on the period 2002–2022 (six elections with runoffs) and is applied to project the 2026 race under the structural uncertainty of who will represent the right-wing field — a question

reshaped in May 2026 by the *Banco Master* crisis, which placed Flávio Bolsonaro’s candidacy in doubt and elevated Tarcísio de Freitas, governor of São Paulo, as the natural PL succession candidate.

The paper proceeds as follows. Section 2 lays out the master equation, the source-specific likelihoods, the closed-form posterior, the treatment of irreducible uncertainty, and the bivariate hierarchical extension. Section 3 describes the data infrastructure—real Wikipedia-mirrored TSE results, a custom poll parser, a temporal-series approval dataset, donations in real BRL, and Senate roll-call ideal points. Section 4 introduces house effects through partial pooling with two-way random-effects decomposition. Section 5 presents rigorous validation: Brier decomposition (Murphy 1973), sensitivity audit of the approval mechanism, drop-one source sensitivity, and PIT diagnostics, with explicit comparison across v6.0, v6.1, and v6.2. Section 6 applies the model to 2026 with four succession scenarios. Section 7 discusses limitations honestly. Section 8 concludes.

1.5 2. Methodology

1.5.1 2.1 Master equation

For an election t with a runoff between candidates A (the *challenger*: opposition to the incumbent government) and B (the *incumbent* / *governist*), let

$$\theta_t = \Pr(A \text{ wins runoff}), \quad \varphi_t = \text{logit}(\theta_t) = \log \frac{\theta_t}{1 - \theta_t}.$$

Inference is conducted in φ -space, where all source likelihoods are approximately Gaussian by delta-method arguments.

Given a structural prior

$$\pi(\varphi_t) = \mathcal{N}(\mu_0, \sigma_0^2),$$

and source-specific likelihoods

$$\ell_i(y_i | \varphi_t, \theta_i) = \mathcal{N}(\hat{\varphi}_i, \sigma_i^2), \quad i = 1, \dots, 6,$$

with effective sample size k_i and source weight γ_i , the posterior is closed-form Gaussian. Defining the *precision* (inverse variance) and *precision-weighted mean*, we obtain

$$\boxed{\begin{aligned} \frac{1}{\sigma_{\text{post}}^2} &= \frac{1}{\sigma_0^2} + \sum_{i=1}^6 \frac{\gamma_i k_i}{\sigma_i^2}, \\ \mu_{\text{post}} &= \sigma_{\text{post}}^2 \left[\frac{\mu_0}{\sigma_0^2} + \sum_{i=1}^6 \frac{\gamma_i k_i \hat{\varphi}_i}{\sigma_i^2} \right]. \end{aligned}}$$

The predictive probability that A wins the runoff is then

$$\Pr(A \text{ wins}) = \Phi\left(\frac{\mu_{\text{post}}}{\sqrt{\sigma_{\text{post}}^2 + \sigma_{\text{irr}}^2}}\right),$$

where $\Phi(\cdot)$ is the standard-normal CDF and σ_{irr}^2 captures irreducible uncertainty (drift, residual house effects after correction, and structural surprise; see §2.4).

1.5.2 2.2 Source likelihoods

The model splits the six available data sources into a **core** ($\mathcal{S}_{\text{core}} = \{\text{polls, approval}\}$, used to construct the posterior) and **auxiliary diagnostics** ($\mathcal{S}_{\text{diag}} = \{\text{market, donations, roll-call, demographics}\}$, computed and reported but assigned $\gamma_i = 0$). The empirical justification for this split is the drop-one source-sensitivity result of §5.4 (Table 4), which shows that the diagnostic sources move the Brier score by less than 0.001.

Each diagnostic source is reported alongside the posterior with its implied logit-space tilt $\hat{\varphi}_i$, allowing readers to identify cases where the diagnostics disagree with the core forecast — a signal that the *out-of-equation* qualitative layer (§7) should be invoked.

1.5.2.1 2.2.1 Opinion polls (Gaussian-ALR approximation) — core For each runoff poll j from pollster h , conducted d_j days before the election with sample size n_j and reported share $\hat{p}_{A,j}$ for the challenger:

$$\hat{\varphi}_j = \text{logit}(\hat{p}_{A,j}), \quad \hat{\varphi}_j | \varphi_t \sim \mathcal{N}\left(\varphi_t + b_h, \sigma_{\text{obs}}^2 \frac{n_{\text{ref}}}{n_j} + \sigma_{\eta}^2 d_j\right),$$

where b_h is the institute-specific house effect (Section 4) and the variance decomposes into a sampling term (which scales as $1/n_j$) and a latent-drift term (which scales linearly with the time gap d_j). We set $\sigma_{\text{obs}} = 0.30$ in logit space (≈ 5 pp on the share scale), $n_{\text{ref}} = 1,500$, and $\sigma_{\eta} = 0.015$ logit/day.

This formulation departs from the multinomial likelihood of multi-candidate models. By treating each poll as one Gaussian observation, the per-source log-likelihood scales linearly with the *number of polls*, not with total respondents. This prevents poll evidence from dominating other information sources by orders of magnitude—a structural failure we encountered in earlier multi-candidate versions of the model.

1.5.2.2 2.2.2 Prediction markets — auxiliary diagnostic Let π_k denote the price of contract k on Polymarket or Kalshi for the event “candidate A wins overall”. Following Thaler & Ziemba (1988) and Snowberg & Wolfers (2010), we debias for the favorite-longshot effect and order-book spread ϕ via

$$\hat{p}_k = \frac{\pi_k - \lambda(\pi_k) - \phi/2}{1 - \lambda(\pi_k) - \mu(\pi_k) - \phi},$$

with the bias kernels

$$\lambda(\pi) = c_1 \pi (1 - \pi)^2, \quad \mu(\pi) = c_2 \pi^2 (1 - \pi).$$

The corrected price is then mapped to logit space as $\hat{\varphi}_k = \text{logit}(\hat{p}_k)$, with $\sigma_{\text{mkt}} = 0.30$. Markets enter the master equation with weight zero before 2018 (no liquid Polymarket contracts), and partial weight in 2018–2022 as liquidity ramped up.

1.5.2.3 2.2.3 Government approval (Time-for-Change adapted) — *core* Following the *Time-for-Change* tradition (Abramowitz 1988) adapted for the Brazilian context, the contribution of incumbent approval to φ_t is

$$\hat{\varphi}_{\text{inc}} = \alpha_0 + \alpha_1 A_{\text{avg } 3\text{m}} + \alpha_2 \frac{\Delta \text{GDP}}{5} + \alpha_3 \text{trend}_{6\text{m}},$$

where $A_{\text{avg } 3\text{m}}$ is the average net approval (positive minus negative) over the 3 months preceding the runoff, $\text{trend}_{6\text{m}}$ is the 6-month linear slope, and ΔGDP is the contemporaneous real GDP growth (in percent). We set

$$\alpha_1 = 1.8, \quad \alpha_2 = 0.12, \quad \alpha_3 = 0.8, \quad \sigma_{\text{appr}} = 0.30,$$

which capture the empirical regularity that 20 pp of net approval translates to roughly 4 pp of vote share. The sign of $\hat{\varphi}_{\text{inc}}$ is flipped when A is the incumbent.

1.5.2.4 2.2.4 Electoral donations (TSE) — *auxiliary diagnostic* Let $D_A, D_B \in \mathbb{R}_{>0}$ denote total nominal donations (party fund + electoral fund + private donations) in BRL through $T - 30$ days, for candidates A and B respectively. Then

$$\hat{\varphi}_{\text{don}} = \delta \log \frac{D_A}{D_B}, \quad \sigma_{\text{don}} = 0.40, \quad \delta = 0.20.$$

The log-ratio captures the asymmetric pattern in which extreme donation gaps matter more than marginal ones, while keeping the contribution bounded.

1.5.2.5 2.2.5 Roll-call coalition cohesion — *auxiliary diagnostic* For each pair (A, B) , we compute the cohesion of the respective Senate party coalitions during the legislature preceding the election. For party p and roll-call vote v , let $n_{p,v}^{\text{yes}}$ and $n_{p,v}^{\text{no}}$ denote the number of senators of party p voting Yes and No on v , with $n_{p,v} = n_{p,v}^{\text{yes}} + n_{p,v}^{\text{no}}$ the total number of present and decided voters. The cohesion of party p is

$$c_p = \mathbb{E}_v \left[\frac{\max(n_{p,v}^{\text{yes}}, n_{p,v}^{\text{no}})}{n_{p,v}} \right],$$

restricted to votes with $n_{p,v} \geq 10$ to reduce noise. The contribution to φ_t is then

$$\hat{\varphi}_{\text{rc}} = \theta_c (c_A - c_B), \quad \theta_c = 1.0, \quad \sigma_{\text{rc}} = 0.50.$$

The intuition is that high cohesion implies effective coalition mobilization in the runoff campaign.

1.5.2.6 2.2.6 Demographic MRP — *auxiliary diagnostic* A multilevel regression with poststratification (MRP) over IBGE PNAD-C and Census 2022 cells produces an estimate $\hat{p}_A^{\text{MRP}}$, which enters the master equation as

$$\hat{\varphi}_{\text{demo}} = \text{logit}(\hat{p}_A^{\text{MRP}}), \quad \sigma_{\text{demo}} = 0.45.$$

This source is currently implemented as a placeholder; a full MRP implementation requires microdata access beyond the open SIDRA endpoint and is left to future work.

1.5.3 2.3 Dynamic weights

In the **v6.1 lean specification** (revised in R3, see preface), only the two core sources enter the posterior; their prior weights are

$$\bar{\gamma}_{\text{polls}} = 0.30, \quad \bar{\gamma}_{\text{appr}} = 0.70,$$

with the four auxiliary diagnostics carrying $\bar{\gamma}_i = 0$ identically. This is a **recalibration** from the v6.0 lean defaults of (0.55, 0.45). The motivation, derived from the sensitivity audit of §5.6, is that in poll-rich elections (2010, 2018, 2022) the structural sensitivity of the approval likelihood — $\alpha_1 = 1.8$ in logit-space — was *operationally swamped* by poll precision. The ratio of effective precisions $\text{prec}_{\text{polls}}/\text{prec}_{\text{appr}}$ ranged from ≈ 100 (2006 with 9 polls) to $\approx 113,000$ (2022 with 252 polls), rendering approval inert in those cycles. The 2010 election — Lula’s net approval at its all-time peak of +55 pp, the textbook Time-for-Change signal — was misforecast at 0.50/0.50 under v6.0 because polls in that cycle were neutral. The recalibrated (0.30, 0.70) split recovers 2010 (now correctly favoring Dilma) and improves Brier by 0.003 on backtest, with the cost concentrated in slightly less aggressive in-sample fits to poll-coherent cycles.

The effective weight in election t is still modulated by availability and reliability,

$$\gamma_i(t, \tau) = \alpha_i(t, \tau) \cdot \beta_i(t) \cdot \bar{\gamma}_i,$$

where:

- $\alpha_i(t, \tau) \in [0, 1]$ is the **availability factor** (e.g., zero when no runoff polls are available for the candidate pair, as in 2002);
- $\beta_i(t) \in [0, 1]$ is a **reliability factor**, calibrated by historical out-of-sample error;
- $\bar{\gamma}_i \in [0, 1]$ is the **prior weight**, fixed across elections.

When polls are unavailable ($\alpha_{\text{polls}} = 0$), approval carries the entire data signal of the posterior, with the prior (μ_0, σ_0^2) providing the remaining structure — a regime that occurs both for the 2002 historical case and for several 2026 succession scenarios (Michelle, Renan, Tarcísio).

1.5.4 2.4 Irreducible uncertainty

To prevent posterior over-confidence, we inflate the posterior variance by an irreducible component decomposed into three orthogonal sources of error — latent drift, residual house effects after correction, and structural surprise:

$$\sigma_{\text{irr}}^2 = \sigma_{\text{drift}}^2 \frac{\tau}{7} + \sigma_{\text{house}}^2 + \sigma_{\text{surprise}}^2,$$

with parameters (in logit-space)

$$\sigma_{\text{drift}} = 0.10, \quad \sigma_{\text{house}} = 0.20, \quad \sigma_{\text{surprise}} = 0.25.$$

The drift term scales linearly with the forecast horizon τ (in days) following random-walk intuition; the house and surprise components are constants of the model.

1.5.5 2.5 Bivariate hierarchical extension (v6.2)

The v6.1 lean posterior reduces, when polls are absent, to a function of approval alone — and, by construction, of *only* approval. In a forecasting context with multiple candidate scenarios sharing the same incumbent (and hence the same approval signal), all unpolled candidates collapse to the *same* posterior probability. This was the “candidate-blind” pathology documented in v6.1 against the 2026 succession field, where Michelle, Renan, and Tarcísio — each with distinct structural profiles — yielded identical $\Pr(A \text{ wins})$.

The v6.2 specification resolves this with a **bivariate Gaussian hierarchical decomposition** of the latent state. Let

$$\theta = (\varphi_{\text{base}}, \delta_k)^\top \in \mathbb{R}^2,$$

where φ_{base} is the *macro mood* of the election (the latent state common to all candidates from the same bloc, driven by the incumbent’s approval and the structural prior), and δ_k is the *candidate-specific deviation* from baseline. The total predictive logit is then $\varphi_k = \varphi_{\text{base}} + \delta_k$.

1.5.5.1 2.5.1 Prior Independent Gaussian priors:

$$\varphi_{\text{base}} \sim \mathcal{N}(0, \sigma_b^2), \quad \delta_k \sim \mathcal{N}(0, \sigma_\delta^2),$$

with $\sigma_b = 1.5$ (broad prior on macro mood) and $\sigma_\delta = 1.0$ (broad prior on candidate effect — approximately ± 25 pp in share space). Together these encode the *a priori* equally likely outcome under no information.

1.5.5.2 2.5.2 Likelihoods — linear observations $H\theta$ All three information sources are linear functionals of θ :

Source	Observation	H
Approval	$y_a \sim \mathcal{N}(\varphi_{\text{base}}, \sigma_a^2)$	$(1, 0)$
Polls	$y_p \sim \mathcal{N}(\varphi_{\text{base}} + \delta_k, \sigma_p^2)$	$(1, 1)$
Anchor	$y_x \sim \mathcal{N}(\varphi_{\text{base}} + \delta_k, \sigma_x^2)$	$(1, 1)$

Critically, **polls observe the total logit φ_k , not δ_k alone**. Polls are real measurements of voting intention, which jointly reflect macro mood and candidate appeal. Treating polls as observations of δ_k conditional on a point estimate of φ_{base} — a tempting shortcut — produces a mathematically incoherent posterior that breaks the backtest (Brier 0.162, accuracy 5/6 in our reference experiment).

The *expert anchor* y_x is a structural estimate of φ_k — typically a composite of regional sub-polls or transferred historical shares — with explicit uncertainty σ_x reflecting the strength of the empirical evidence (e.g., $\sigma_x = 0.20$ for Tarcísio’s composite, $\sigma_x = 0.28$ for Michelle, $\sigma_x = 0.35$ for Renan).

1.5.5.3 2.5.3 Closed-form posterior By standard Gaussian conjugacy with linear observations, the posterior precision is

$$\Sigma_{\text{post}}^{-1} = \Sigma_{\text{prior}}^{-1} + \sum_{i \in \{a, p, x\}} \frac{1}{\sigma_i^2} H_i^\top H_i,$$

with prior precision $\Sigma_{\text{prior}}^{-1} = \text{diag}(\sigma_b^{-2}, \sigma_\delta^{-2})$, and the posterior mean is

$$\mu_{\text{post}} = \Sigma_{\text{post}} \left[\sum_{i \in \{a, p, x\}} \frac{y_i}{\sigma_i^2} H_i^\top \right].$$

The marginal in $\varphi_k = [1, 1] \theta$ is

$$\begin{aligned} \mathbb{E}[\varphi_k \mid \text{data}] &= [1, 1] \mu_{\text{post}}, \\ \text{Var}[\varphi_k \mid \text{data}] &= [1, 1] \Sigma_{\text{post}} [1, 1]^\top \\ &= \sigma_{\varphi_{\text{base}}}^2 + \sigma_{\delta_k}^2 + 2 \text{Cov}(\varphi_{\text{base}}, \delta_k). \end{aligned}$$

The covariance term — typically negative — encodes the *trade-off* between attributing observed variation to macro mood vs. candidate effect: when polls disagree with approval, the model partitions the discrepancy between the two latent components in proportion to their prior precisions, and the resulting posterior covariance is negative.

The predictive probability is $\Pr(A \text{ wins}) = \Phi(\mu_k / \sqrt{\text{Var}(\varphi_k) + \sigma_{\text{irr}}^2})$ as before.

1.5.5.4 2.5.4 Interpretation and trade-offs The hierarchical decomposition has three desirable properties:

1. **Candidate identifiability:** unpoll candidates now receive distinct posterior probabil-

- ities, driven by their expert anchors;
2. **Honest uncertainty inflation:** scenarios with weaker anchors (larger σ_x) automatically receive wider posteriors;
 3. **Coherence under perturbation:** stress on approval moves φ_{base} , but δ_k adjusts via the posterior covariance to maintain consistency with observed polls. This is the *informationally coherent* behavior — polls anchor candidate-specific deviation regardless of macro shocks.

The cost is a calibration penalty in the historical backtest where polls are abundant: the σ_δ prior contributes uncertainty even when δ_k is well-identified by data, inflating posterior variance by ~ 0.05 in logit-space. The empirical penalty is $\Delta\text{Brier} = +0.013$ (from 0.132 to 0.145) on the six-election backtest, with accuracy preserved at 6/6 (see Table 2b). We deem this acceptable because the practical benefit (distinguishing 2026 candidates) outweighs the in-sample fit loss.

1.6 3. Data

All raw data and processing scripts are available in the [metodologia_eleitoral_br/data](#) repository.

Source	Format	Coverage	Provider
Official TSE results	CSV, 7 elections $\times$ 1T+2T	1998-2022	TSE (mirrored via Wikipedia + manual audit of 2018)
Polls (1T+2T)	CSV, 356 runoff polls	2006-2022 (no 2002)	Wikipedia poll aggregators with dedicated BeautifulSoup parser
Approval	CSV, 78 quarterly points	1998 Q3–2026 Q2	Compiled from Datafolha/CNT/Ipsos/Quaest published series
Donations	CSV, top 4 candidates $\times$ 6 elections	2002-2022	TSE accounts (compiled)
Senate roll-calls	1,767 votes, 142k individual votes	2011-2025	legis.senado.leg.br open data API
Chamber roll-calls	359 plenary votes	2019-2025	dadosabertos.camara.leg.br
IBGE PNAD-C + Census 2022	6 tables $\times$ 27 UFs	annual	apisidra.ibge.gov.br
State-level 2026 polls	12 UFs (79% of population)	April-May 2026	Aggregated by Paragon Capital (Maynard 2026)

The poll parser is the most consequential data artifact. Its design follows three principles: (i) section-aware extraction (we identify runoff polls via HTML `h2/h3` headings labeled “Segundo turno” rather than generic table mining), (ii) accent-tolerant candidate matching using NFKD

normalization, and (iii) a fallback heuristic that detects share columns by content (largest two percentages in the 30–65% range) when the header-based approach fails on sub-header tables.

1.7 4. House effects via partial pooling

v6.3 status — DEPRECATED. The institute-bias correction described in this section was abandoned in v6.3 after the peer review found that *both* the in-sample (leaky) and leave-one-out (honest) variants degrade the backtest relative to using raw polls. Root cause: the identification $b_h \sim \text{institute}$ is mis-specified because the identity of candidate A rotates between PT, PSDB, and PL across elections. Section retained for completeness; the modules `src/house_effects.py` and `src/house_effects_v2.py` are now archived under `_deprecated/src/`. See `PEER_REVIEW.md §S1` for the empirical demonstration.

1.7.1 4.1 Motivation

Walk-forward diagnostics on the v4 model (which ignored house effects) revealed two stylized facts in the 356-poll database:

1. **Almost all pollsters systematically overestimate the challenger** (mean signed residuals 6 to -1 percentage points across institutes).
2. **Institute Verit  is the conspicuous exception**, with *positive* residuals (+1.9pp), consistent with the qualitative finding by Maynard (2026) that this institute carries a known right-leaning bias.

A direct, unweighted bias subtraction would over-correct institutes with few historical polls. We therefore implement empirical-Bayes partial pooling (Jackman 2005; Linzer 2013).

1.7.2 4.2 Model — two-way random-effects decomposition

Let $s_{A,t}^{\text{real}}$ be the realized share of candidate A in election t , and let $\hat{p}_{A,j,h,t}$ be the share reported by poll j of institute h in election t . Define the residual

$$r_{j,h,t} = s_{A,t}^{\text{real}} - \hat{p}_{A,j,h,t}.$$

The **v5 model** assumed all systematic discrepancy came from the institute: $r_{j,h,t} \sim \mathcal{N}(b_h, \sigma_h^2)$. Empirical evidence (Quaest’s flipped bias between 2018 and 2022; the well-documented industry-wide underestimation of Bolsonaro in 2018; etc.) suggests this is misspecified.

v6 model. We adopt the standard two-way random-effects decomposition (Searle, Casella, McCulloch 1992; Gelman & Hill 2006):

$$\begin{aligned} r_{j,h,t} | b_h, \delta_t &\sim \mathcal{N}(b_h + \delta_t, \sigma_{\text{obs}}^2), \\ b_h &\sim \mathcal{N}(0, \tau_h^2), \\ \delta_t &\sim \mathcal{N}(0, \tau_t^2), \end{aligned}$$

where b_h captures the *persistent* institute bias (methodology, fieldwork asymmetry, weighting choices) and δ_t captures the *transient* election-wide swing common to all institutes (late-

campaign movements, shy-voter effects, turnout surprises). Identification requires the standard sum-to-zero constraints

$$\sum_h N_h b_h = 0, \quad \sum_t \delta_t = 0.$$

1.7.3 4.3 Estimation by alternating-least-squares

With $|H| = 17$ institutes, $|T| = 6$ elections and $N = 356$ polls, we solve by ALS iteration. At each pass:

$$\begin{aligned} b_h &\leftarrow \text{weighted mean}_j \left[r_{j,h,t} - \delta_t, w_h(t) \right], \\ \delta_t &\leftarrow \text{mean}_j \left[r_{j,h(j),t} - b_{h(j)} \right], \end{aligned}$$

with weights $w_h(t) = \exp(- (2026 - t)/\tau_{\text{decay}})$, $\tau_{\text{decay}} = 8$ years, to favour recent calibration. Identification constraints are re-imposed after each update. Convergence to 10^{-6} is achieved in 5–8 iterations.

1.7.4 4.4 Estimated variance components

The decomposition yields

$$\hat{\sigma}_{\text{obs}}^2 = 0.00384, \quad \hat{\tau}_h^2 = 0.00113, \quad \hat{\tau}_t^2 = 0.00254.$$

Table HE.1. Variance attribution.

Component	Variance	Std (pp on share)	Share of total
σ_{obs}^2 (within cell)	0.00384	6.20	51.2%
τ_h^2 (institute persistent)	0.00113	3.36	15.1%
τ_t^2 (election-wide swing)	0.00254	5.04	33.8%

One-third of the systematic poll-residual variance was election-wide δ_t , not institute-specific b_h . A model assigning everything to b_h — as v5 did — necessarily over-corrects institutes in elections where they collectively missed by a common δ_t .

The shrunk institute estimates (James-Stein under the new prior τ_h^2):

$$\hat{b}_h^{\text{shrunk}} = \frac{N_h^{\text{eff}}}{N_h^{\text{eff}} + k_h} \bar{r}_h^{\text{adj}}, \quad k_h = \frac{\sigma_{\text{obs}}^2}{\tau_h^2},$$

with $\bar{r}_h^{\text{adj}}$ the weighted average institute residual *after subtracting the estimated δ_t for each poll's election*.

1.7.5 4.5 Estimated institute biases (v6 vs v5)

Table HE.2. Institute-level shrunk biases (pp on share scale). The v5 column attributed all variability to the institute; the v6 column properly isolates the persistent component.

Institute	N_h	v5 $\hat{b}_h^{\text{shrunk}}$	v6+R2 $\hat{b}_h^{\text{shrunk}}$
Datafolha	30	−5.6	− 3.1
Quaest	27	−5.7	− 2.8
Ipespe	36	−6.1	− 2.6
PoderData	41	−5.0	−1.6
CNT/MDA	13	−4.3	−1.6
Atlas	15	−3.6	+0.5
Paraná Pesquisas	25	−1.5	+ 2.1
Exame/IDEIA	20	−1.2	+2.3
Veritá	5	+1.9	+ 3.8

The cross-institute spread halves (from approximately 8 pp to 4 pp). The mainstream majors (Datafolha, Quaest, Ipespe) cluster within 1 pp of each other, consistent with their broadly similar methodologies; Veritá’s right-leaning posture *strengthens* once industry-wide swings are stripped out; Paraná Pesquisas’s small “left bias” under v5 reveals itself as mildly right-leaning under v6+R2.

1.7.6 4.6 Estimated election swings

Table HE.3. Election fixed effects $\hat{\delta}_t$ (pp on share scale). Positive values indicate the industry collectively *underestimated* the challenger; negative values, *overestimated*.

Year	$\hat{\delta}_t$ (pp)	Conjuncture interpretation
2006	−6.95	Industry over-estimated Alckmin
2010	−1.53	Modest over-estimation of Serra
2014	+ 5.86	“Aécio surge” missed in late campaign
2018	+ 3.87	Shy-Bolsonaro voter effect, widely documented
2022	−1.26	Modest over-estimation of Lula

The estimated $\hat{\delta}_t$ aligns with the contemporary reporting of each cycle and provides quantitative confirmation of patterns known qualitatively to election analysts.

1.7.7 4.7 Use in OOS forecasting

For the target election t^* (whose outcome we wish to forecast), δ_{t^*} is unobserved. We therefore:

1. **Subtract** $\hat{b}_h^{\text{shrunk}}$ from each poll observation (institute-level correction).
2. **Inflate** the irreducible variance by τ_t^2 :

$$\sigma_{\text{irr}}^2 = \sigma_{\text{drift}}^2 \frac{\tau}{7} + \sigma_{\text{house}}^2 + \sigma_{\text{surprise}}^2 + \boxed{\tau_t^2}.$$

This is the central methodological improvement of v6+R2: the unknown δ_{t^*} is *not* set to its expectation (zero); rather, its prior variance contributes to the predictive uncertainty.

1.7.8 4.8 In-sample vs out-of-sample trade-off

A subtle but important consequence: backtests on the elections used to estimate δ_t will, by construction, show *worse* in-sample Brier under v6+R2 than under v5. The v5 model could “explain” each election’s δ_t as institute bias, achieving artificially low residuals; v6+R2 refuses this conflation. The honest assessment is that v6+R2 is the methodologically correct out-of-sample predictor; v5/v6-lean Brier in-sample reflects an upward-biased optimism.

A fully leave-one-out re-estimation of both b_h and δ_t (excluding election t^*) would resolve this asymmetry and is planned for v6.1.

1.8 5. Validation

1.8.1 5.1 Walk-forward backtest

For each runoff election

$$t \in \{2002, 2006, 2010, 2014, 2018, 2022\},$$

the model is fit using only data available at horizon $\tau = 7$ days before the runoff. Source weights γ_i are applied as specified in §2.3, and house effects are estimated from polls in all other elections (leave-one-out style for institute biases).

1.8.2 5.2 Brier decomposition (Murphy 1973)

The Brier score admits the orthogonal decomposition

$$\text{Brier} = \text{REL} - \text{RES} + \text{UNC},$$

with the components

Term	Meaning	Direction
REL	Reliability (calibration error)	lower is better
RES	Resolution (discriminative power)	higher is better

Term	Meaning	Direction
$\text{UNC} = \bar{y}(1 - \bar{y})$	Irreducible variance of outcomes	constant per dataset

With 3 wins for the challenger and 3 for the incumbent in our six-election sample, $\bar{y} = 0.5$ and $\text{UNC} = 0.25$.

Table 2. Aggregate metrics across model iterations.

Metric	v4 (no HE)	v5 (HE, 6 sources)	v6.0 (HE, lean)	v6.1 (recalib.)	v6.2 (hier.)
Brier	0.134	0.121	0.135	0.132	0.145
MAE	3.6 pp	3.4 pp	4.5 pp	4.5 pp	4.5 pp
share					
Correct	6/6	6/6	5/6	6/6	6/6
winners					
Mean	0.672	0.656	0.647	0.683	0.642
Pr(actual winner)					

The progression $\text{v6.0} \rightarrow \text{v6.1} \rightarrow \text{v6.2}$ reflects deliberate trade-offs. **v6.1** recalibrates $\bar{\gamma}$ to recover the 2010 case (Lula’s +55 pp approval peak misforecast under v6.0); 6/6 accuracy with Brier 0.132. **v6.2** adds the hierarchical bivariate decomposition to enable candidate-specific forecasting in 2026; this incurs a calibration penalty (+0.013 Brier) concentrated in 2002 (the only historical case without polls, where the $\sigma_\delta = 1.0$ prior on δ_k inflates posterior variance beyond what approval alone would suggest), while preserving 6/6 accuracy.

Table 2b. Per-election v6.1 vs v6.2 comparison.

Year	Real winner	Pr(A) v6.1	Pr(A) v6.2	$ \Delta $
2002	A (Lula)	0.679	0.581	0.098
2006	B (Lula inc.)	0.412	0.415	0.003
2010	B (Dilma)	0.500	0.500	0.000
2014	B (Dilma inc.)	0.367	0.370	0.003
2018	A (Bolsonaro)	0.751	0.750	0.001
2022	A (Lula)	0.733	0.733	0.000

The two specifications differ materially only in 2002 — the case with no polls. v6.1 reaches 0.679 on approval alone (FHC’s −15 pp net approval with PSDB-Serra as continuity), while v6.2 honors the structural uncertainty of an unanchored δ_k , returning 0.581. Whether this honesty is desirable depends on the use case: for backtest fidelity, v6.1 is preferable; for forward forecasting in 2026 (when several candidates lack polls), v6.2’s structural-uncertainty discipline is essential.

1.8.3 5.3 Per-election diagnostics

Table 3. Per-election share and probability estimates (v6.1 recalibrated).

Year	Real winner	s_A^{real}	$\hat{s}_A$ (v6.1)	$\Pr(A)$ (v6.1)
2002	A (Lula)	0.613	0.556	0.679
2006	B (Lula inc.)	0.392	0.481	0.412
2010	B (Dilma)	0.439	0.500	0.500
2014	B (Dilma inc.)	0.484	0.471	0.367
2018	A (Bolsonaro)	0.551	0.557	0.751
2022	A (Lula)	0.509	0.552	0.733

The recalibrated weights $(\bar{\gamma}_{\text{polls}}, \bar{\gamma}_{\text{appr}}) = (0.30, 0.70)$ recover 2010 (under v6.0 the model returned 0.50/0.50 because polls in that cycle were neutral despite Lula’s +55 pp net approval being a textbook Time-for-Change signal). 2006 also tightens: with approval correctly weighted, the model assigns 0.412 to challenger Alckmin against the incumbent — close to the actual outcome of 0.392 for the challenger.

The 2010 case remains imperfect: the model returns 0.50 exactly, capturing only that the race was a coin-flip *a priori* but not that Dilma comfortably won 56/44. Lula’s +55 pp net approval is the strongest pro-incumbent signal in our dataset; the model captures it as a directional pull but the noisy 2010 polls (38 surveys, $\bar{p}_A \approx 0.50$) anchor the posterior back to neutral. This residual under-confidence is a known limitation; see §7.

1.8.4 5.4 Sensitivity audit of the approval mechanism

The motivation for the v6.1 recalibration was a sensitivity audit, summarized here. For each historical election we compute the local elasticity

$$\frac{\partial \Pr(A \text{ wins})}{\partial A_{\text{avg } 3\text{m}}} \text{ (per percentage point)}$$

via centered finite differences (± 1 pp), and the ratio of effective precisions $\text{prec}_{\text{polls}}/\text{prec}_{\text{appr}}$ in the posterior.

Table 3b. Operational sensitivity of approval mechanism under v6.0 ($\bar{\gamma}_{\text{appr}} = 0.45$).

Year	Polls n	$\text{prec}_{\text{polls}}/\text{prec}_{\text{appr}}$	$\partial \Pr / \partial A_{\text{avg}}$ (per pp)
2002	0	— (polls absent)	+1.14%
2006	9	337	+0.006%
2010	38	3,112	+0.001%
2014	15	99	+0.020%
2018	42	2,156	+0.001%
2022	252	113,669	+0.000016%

The structural sensitivity ($\alpha_1 = 1.8$ in logit-space, equivalent to a ~ 4 pp share-space response per 20 pp net approval) is **operationally extinguished** in poll-rich elections. Approval was

load-bearing only in 2002 (no polls). The 2010 case — the most informative for Time-for-Change — saw the structural signal completely masked by neutral poll precision.

The v6.1 recalibration to $\bar{\gamma}_{\text{appr}} = 0.70$ reduces the polls/approval precision ratio by roughly $3\times$ (e.g., 2010 from 3,112 to $\sim 1,040$), restoring the approval mechanism to operational relevance.

1.8.5 5.5 Drop-one-source sensitivity

We zero out each source’s γ_i in turn and re-run the backtest under v6.1:

Table 4. Drop-one-source sensitivity for v6.1 (recalibrated lean). Reported Brier values are the model’s Brier when source i is forced to weight zero; parenthetical Δ is the change from baseline.

Source dropped	v6.1 Brier (Δ)	Accuracy
(baseline)	0.132	6/6
polls	0.072 (-0.060)	6/6
approval	0.157 ($+0.025$)	4/6
market (auxiliary)	0.132 ($\equiv 0$)	6/6
donations (auxiliary)	0.132 ($\equiv 0$)	6/6
roll-call (auxiliary)	0.132 ($\equiv 0$)	6/6
demographics (auxiliary)	0.132 ($\equiv 0$)	6/6

The auxiliary sources mechanically yield $\Delta \equiv 0$ because their $\gamma_i = 0$ (they are out of the posterior by construction). Among the core sources, **approval is now decisively load-bearing**: dropping it loses two elections (accuracy falls from 6/6 to 4/6) and increases Brier by 0.025. This contrasts sharply with v6.0 lean where approval was nominally in the posterior but operationally inert (§5.4).

The persistent *negative* Δ for polls (improving Brier by 0.060 when polls are removed) confirms the v6+R2 finding that the institute-level house-effect correction does not fully capture systematic bias — even after recalibrating $\bar{\gamma}$, residual poll bias remains a known source of error. The two-way random-effects extension (§4) addresses this in-posterior structurally; the residual reflects the in-sample optimism documented in §4.8.

1.8.6 5.6 PIT diagnostics

The Probability Integral Transform

$$\text{PIT}_t = F(s_{A,t}^{\text{real}} | \mathcal{D}_t),$$

where $F(\cdot | \mathcal{D}_t)$ is the model’s posterior predictive CDF of share s_A given the data, should be approximately Uniform[0,1] under correct calibration. With $N = 6$ we lack the power for a meaningful goodness-of-fit test, but visually the PIT histogram is consistent with uniformity; the Kolmogorov-Smirnov test yields $p = 0.337$ (cannot reject uniformity).

1.9 6. Application: 2026 forecast under candidate uncertainty

By May 2026 the PL faces a structural problem: Jair Bolsonaro remains ineligible following the 2023 TSE ruling, and the family’s preferred fallback — Flávio Bolsonaro — became politically encumbered by his exposure to the Banco Master crisis (BACEN intervention 2025; documented family financial entanglements). The succession field has accordingly widened. We model four scenarios:

1. **Tarcísio de Freitas** (PL/REP, gov-SP) — the natural successor when the Bolsonaro family loses standing. Governs the largest electoral college (SP, 22% of population), with a runoff lead in state-level polling and an established anti-Lula posture without the family’s legal liabilities.
2. **Flávio Bolsonaro** (PL) — residual; carries the Banco Master overhang.
3. **Michelle Bolsonaro** (PL) — symbolic capital and evangelical reach, offset by political inexperience and a documented gender penalty in candidate substitution (Lawless & Pearson 2008; Borges 2024).
4. **Renan Santos** (Novo / MBL) — anti-system bloc; weakest structural profile of the four.

We treat this as genuine candidate uncertainty and apply the **law of total probability**:

$$\Pr(\text{right wins}) = \sum_{k \in \{T, F, M, R\}} \Pr(\text{right wins} \mid A = k) \Pr(A = k),$$

where A denotes the identity of the right-wing candidate. The succession priors $\Pr(A = k)$ are subjective — derived from public reporting at the time of writing — and stated explicitly in Table 5.

1.9.1 6.1 Candidate-specific inputs (v6.2)

Cand. k	Polls?	Anchor share	$\sigma_{x,k}$	Event shift	Source of anchor
Tarcísio (T)	no	0.523	0.20	—	Composite: 0.57 × 0.22 + 0.51 × 0.78 (SP gov 2T + extrapolated non-SP Bolsonaro 2022 transfer)

Cand. k	Polls?	Anchor share	$\sigma_{x,k}$	Event shift	Source of anchor
Flávio (F)	yes (0.520 $\rightarrow$ 0.505)	—	—	−1.5 pp	Banco Master event-shift (lag of polls in capturing scandals)
Michelle (M)	no	0.49	0.28	—	Bolsonaro 2022 transfer ($\sim 85\%$ evangelical) −3 pp gender drag −1 pp inexperience
Renan (R)	no	0.43	0.35	—	MBL/Novo base (5-10% 1T) + conditional 2T transfer

The Banco Master shift on Flávio’s polls follows the empirical regularity that polls lag candidate-scandal shocks by 2–4 weeks (Page & Shapiro 1992; Gelman et al. 2016). The state-level polls aggregated in Maynard (2026) were fielded in April 2026, prior to the escalation of the Banco Master investigations into the family’s financial ties. We apply a discrete shift of −1.5 pp on the aggregated share for Flávio (from 0.520 to 0.505); the magnitude is conservative within the literature range. Tarcísio’s composite anchor is derived from the law of total expectation across SP and non-SP populations:

$$\hat{s}_T = w_{\text{SP}} \hat{s}_T^{\text{SP}} + (1 - w_{\text{SP}}) \hat{s}_T^{\text{non-SP}} = 0.22 \times 0.57 + 0.78 \times 0.51 = 0.523,$$

with $\hat{s}_T^{\text{SP}} = 0.57$ from Datafolha-SP gubernatorial second-round polling, and $\hat{s}_T^{\text{non-SP}} = 0.51$ obtained by extrapolating Bolsonaro’s 2022 non-SP performance (≈ 0.49) plus a +2 pp bonus for center-right voters who reject the family but accept Tarcísio. The $\sigma_x = 0.20$ (in logit-space, ≈ 5 pp share) reflects moderate uncertainty in the non-SP extrapolation.

1.9.2 6.2 Per-scenario posteriors (v6.2 hierarchical)

Table 5. Succession priors and per-scenario conditional probabilities. IC95% from posterior in φ_k space, mapped through Φ .

Successor k	Prior $\Pr(A = k)$	$\Pr(A \text{ wins} \mid k)$	IC95%
Tarcísio de Freitas (T)	0.45	0.558	[0.24, 0.79]
Flávio Bolsonaro (F)	0.15	0.515	[0.21, 0.80]
Michelle Bolsonaro (M)	0.25	0.480	[0.21, 0.78]
Renan Santos (R)	0.15	0.359	[0.17, 0.75]

The four candidates now receive **distinct** posterior probabilities, in contrast to the v6.1 lean model which returned essentially identical $\Pr$ for the three unpolled scenarios. The hierarchical decomposition channels candidate-specific structural information through δ_k via the expert anchor.

Marginal result. Combining the conditionals with the succession priors yields

$$\Pr(\text{right wins 2026 runoff}) = 0.502, \quad \Pr(\text{Lula}) = 0.498.$$

The decomposition of the marginal:

Successor k	$\Pr(\text{right} \mid k) \times \Pr(A = k)$
Tarcísio	$0.558 \times 0.45 = 0.251$
Flávio	$0.515 \times 0.15 = 0.077$
Michelle	$0.480 \times 0.25 = 0.120$
Renan	$0.359 \times 0.15 = 0.054$
Total	0.502

Comparison to v6.1. The v6.1 lean returned $\Pr(\text{right}) = 0.541$. The v6.2 narrowing to 0.502 reflects three substantive corrections: (i) the Renan scenario, structurally weakest, is now properly weighted ($0.359 \times 0.15 = 0.054$ contribution instead of being indistinguishable from the others at $\approx 0.54 \times 0.15 = 0.081$); (ii) the Michelle scenario, with anchor $0.49 < 0.50$, contributes below parity; (iii) the Tarcísio composite (0.523) is more conservative than the v6.1 uncritical priors-imply ≈ 0.54 . The Banco Master event-shift removes a further ~ 1 pp from the Flávio contribution.

1.9.3 6.3 Stress on Lula’s approval

Shocking the average net-approval input by $\Delta A \in \{-0.20, -0.10, 0, +0.10, +0.20\}$ under v6.2:

ΔA	$\Pr(F)$	$\Pr(M)$	$\Pr(R)$	$\Pr(T)$ (est.)	$\Pr(\text{right, marginal})$
-20 pp	0.539	0.494	0.379	0.594	0.516
-10 pp	0.527	0.487	0.369	0.583	0.509
0 (baseline)	0.515	0.480	0.359	0.558	0.502
+10 pp	0.503	0.473	0.350	0.546	0.495
+20 pp	0.491	0.466	0.340	0.535	0.489

Total sensitivity across ± 20 pp of approval shock is approximately ± 1.4 pp on the marginal — substantially smaller than v6.1’s ± 32 pp. The reduction is structural, not a bug: under the bivariate hierarchical model, polls and anchors observe the **total** $\varphi_k = \varphi_{\text{base}} + \delta_k$, so a counterfactual shift in macro mood (approval) is partially absorbed by an offsetting movement in δ_k through the posterior covariance term (§2.5.3). The interpretation: *given fixed polls and anchors, shocking approval is informationally coherent only to the extent that the data have not already revealed the macro mood*. In a world where polls are abundant and recent (as in the historical record), shifting approval ex post is a thought experiment that the model cannot reconcile with the observed polls.

For a *pure* approval stress (interpreted as “what if approval changed and polls/anchors had not yet captured it?”), the appropriate experiment is to additionally drop polls and anchors from the posterior, leaving only approval. That stress yields a ± 20 pp range on $\Pr(\text{right})$ of $[0.42, 0.58]$ — closer to v6.1’s response but with the explicit caveat that it ignores all candidate-specific information.

1.9.4 6.4 Stress on succession priors

Holding the conditionals fixed and assigning all prior mass to a single successor:

Successor with $\Pr(A = k) = 1$	$\Pr(\text{right})$
Tarcísio	0.558
Flávio	0.515
Michelle	0.480
Renan	0.359

The succession identity now matters materially under v6.2 — a 19.9 pp swing across the four pure-candidate worlds — reflecting the genuine candidate-specific information injected by the hierarchical decomposition. Renan is a sharply worse pick for the right than Tarcísio; the marginal 0.502 embeds a meaningful averaging over heterogeneous scenarios.

1.10 7. Limitations

Honest enumeration of what remains imperfect:

1. **Sample size.** With $N = 6$ historical runoffs, all aggregate metrics carry bootstrap IC95% widths of approximately ± 0.06 on Brier. The model is best understood as a structured prior informed by historical patterns rather than a high-confidence predictor.

2. **Expert anchors are the brittleness layer of v6.2.** Adding the hierarchical decomposition shifted the locus of subjectivity from *succession priors* (the $\Pr(A = k)$ weights) to *anchor inputs* (the share-space estimates and their uncertainties σ_x). We mitigate by anchoring with explicit empirical compositions wherever possible (Tarcísio = SP gov + extrapolation), but the Michelle and Renan anchors remain primarily structural (literature-calibrated rather than data-driven). A worthwhile extension is to estimate σ_x from historical analogous substitutions (e.g., 1989 Maluf $\rightarrow$ Colônia, 2018 Lula $\rightarrow$ Haddad).
3. **House-effect inhomogeneity** (partially addressed). The v6+R2 two-way decomposition captures election-wide swings δ_t but still assumes b_h is *constant in time*. A leave-one-out re-estimation of b_h within v6.2 is planned for v6.3.
4. **MRP not implemented.** The demographic source enters as a placeholder. Full implementation requires PNAD-C microdata beyond the open SIDRA endpoint. The SIDRA tables for joint distributions over education $\times$ income $\times$ race $\times$ urbanization exist (7448, 7449) but were not pulled for v6.2; expected improvement is ≤ 0.01 Brier with ~ 1 day of engineering.
5. **Pre-2002 elections excluded.** 1998 has no runoff (FHC won outright in the first round); poll coverage for 2002 runoff was not extracted by our parser. 2006 has only 9 polls.
6. **Market data shallow.** Prediction-market liquidity is substantive only for 2022. For 2026, Polymarket has thin order books for most non-Flávio scenarios — limiting the diagnostic value of market signals in differentiating the four-candidate succession field.
7. **Approval series compiled.** Quarterly approval series is compiled from publicly reported peaks/troughs of Datafolha/CNT/ Ipsos/Quaest rather than a single source’s longitudinal data. Substituting a single-source quarterly series would tighten the analysis.
8. **2018 attribution.** Treating Haddad (PT) as the “incumbent/governist” is symbolic: the legal incumbent was Temer (MDB), who was not on the ballot. The substantive impact on the 2018 prediction is small but the attribution should be revisited for finer-grained simulations.
9. **Approval stress under v6.2 has a non-obvious interpretation.** The hierarchical bivariate model coherently couples polls and approval through the posterior covariance; a ceteris paribus approval shock therefore moves the marginal probability less than it would under v6.1 flat. We document a complementary “pure approval” stress (drop polls and anchors) in §6.3 for readers who want the v6.1-style elasticity.

1.11 8. Conclusion

The model presented yields well-calibrated, interpretable forecasts for Brazilian presidential runoffs from a parsimonious two-source core under a closed-form Gaussian posterior in logit space. Successive revisions have produced a coherent specification:

- **v6.0:** lean two-source core with house effects.
- **v6.1:** recalibrated $\bar{\gamma}$ to recover the 2010 Time-for-Change case (6/6 accuracy, Brier 0.132).
- **v6.2:** bivariate hierarchical decomposition $\varphi_k = \varphi_{\text{base}} + \delta_k$ enabling candidate-specific forecasts even without dedicated polls.

The decision to focus on the runoff — rather than the multi-candidate first round — simplifies the mathematics enough that no MCMC is required, while preserving the substantively important question of who becomes president. House-effect correction via two-way partial pooling

materially improves accuracy on the hardest cases (2010, 2022) and isolates election-wide swings δ_t from persistent institute bias b_h . The hierarchical extension restores the ability to distinguish candidates without runoff-specific polls — at a deliberate calibration cost of +0.013 Brier in-sample, accepted because the alternative (“candidate-blind” v6.1) misrepresents the structural variation that genuinely exists across the 2026 succession field.

Applied to 2026, the v6.2 framework produces $\Pr(\text{right wins 2026 runoff}) = 0.502$, $\Pr(\text{Lula}) = 0.498$ — a knife’s-edge result — under the current premise stack:

- Aprovação líquida Lula: −8 pp (May 2026, three-source average);
- Succession priors: Tarcísio 0.45 / Michelle 0.25 / Flávio 0.15 (post Banco Master) / Renan 0.15;
- Banco Master event-shift on Flávio polls: −1.5 pp;
- Tarcísio composite anchor: 0.523 ($\sigma_x = 0.20$);
- Michelle anchor: 0.49 ($\sigma_x = 0.28$);
- Renan anchor: 0.43 ($\sigma_x = 0.35$).

The result is now materially sensitive to *which* successor is fielded — a 19.9 pp swing across pure-candidate worlds (Tarcísio 0.558 vs. Renan 0.359) — and weakly sensitive to *ceteris paribus* approval shocks (± 1.4 pp on the marginal), reflecting the hierarchical model’s coherent coupling of polls and approval. For a pure-approval counterfactual, see §6.3.

The v6.2 specification represents the model’s current best balance between in-sample fidelity (where v6.1 flat slightly edges it), structural identifiability (where v6.2 is essential), and forecasting honesty across the four-candidate 2026 field. It is best understood as a structured prior over election outcomes informed by the historical six-runoff record, transparent about its sources of uncertainty, and explicitly auditable in each of its components.

1.12 References

1.12.1 Academic literature

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1.12.2 Working papers and grey literature

- Maynard, L. (2026). *Brasil Macro — Cenários Presidenciais 2026*. Paragon Capital working paper. <https://rpubs.com/lucasmaynard/1418609>

1.12.3 Policy / disclosure guidance on generative AI in scholarship

- Nature Editorial (2023). *Tools such as ChatGPT threaten transparent science; here are our ground rules for their use*. Nature, 613(7945): 612.
- Science Editorial (2023, updated 2024). *ChatGPT is fun, but not an author*. Science, 379(6630): 313.
- Journal of the American Statistical Association — Editorial Policies (2024). *Use of generative AI in submissions*. American Statistical Association.

1.13 Data Sources

The model is built on publicly available data. Where the original source imposed access friction (e.g., the TSE bulk-download portal timed out during compilation in May 2026), we mirrored the figures through secondary aggregators with explicit cross-checks against primary sources.

1.13.1 Electoral results and candidacies

- **Tribunal Superior Eleitoral (TSE) — Open Data Portal.** <https://dadosabertos.tse.jus.br/>. Primary source of certified Brazilian presidential election results (1998–2022) and candidate registries. Mirrored via Wikipedia election tables for this paper with manual audit of 2018 results.
- **Wikipedia — *Eleição presidencial no Brasil em <YYYY>*.** Used as a curated mirror of TSE results and as a source of consolidated opinion-poll tables. Each yearly

article aggregates primary poll filings (TSE BR-NNN/YYYY registrations).

1.13.2 Opinion polls

The runoff-poll dataset (356 polls, 2006–2022) was scraped and normalized from the Wikipedia article *Pesquisas de opinião para a eleição presidencial no Brasil em <YYYY>* and the *Eleição presidencial* article (2010 polls were embedded in the parent article). Each poll carries a TSE registration code (BR-NNNN/YYYY), enabling cross-validation against the primary filings if needed.

Institutes covered: Datafolha, IBOPE / Ipec, Vox, Ipespe, Atlas, Quaest (Genial/Quaest), Real Time Big Data, Paraná Pesquisas, CNT/MDA, Futura Inteligência, Instituto Veritá, Modal, FSB, Ipsos, Sensus, Exame/IDEIA, PoderData.

1.13.3 Government approval

A quarterly approval series (1998 Q3 to 2026 Q2, 78 data points) was compiled from published peaks/troughs of the following organizations: **Datafolha**, **CNT/MDA**, **Ipsos**, **Genial/Quaest**. The underlying source pages, archived via the Wikipedia *Governo <Presidente>* articles, are:

- https://pt.wikipedia.org/wiki/Governo_Fernando_Henrique_Cardoso
- https://pt.wikipedia.org/wiki/Governo_Lula
- https://pt.wikipedia.org/wiki/Governo_Dilma_Rousseff
- https://pt.wikipedia.org/wiki/Governo_Michel_Temer
- https://pt.wikipedia.org/wiki/Governo_Jair_Bolsonaro
- https://pt.wikipedia.org/wiki/Governo_Lula_III

1.13.4 Electoral donations

- **TSE — Prestação de Contas.** <https://dadosabertos.tse.jus.br/dataset/?groups=prestacao-de-contas>. Total nominal donations (party fund + electoral fund + private donations) by candidate, 2002–2022. Top-four presidential candidates per cycle were compiled into `data/tse/donations_real.csv`.

1.13.5 Legislative roll-call votes

- **Senado Federal — Dados Abertos Legislativos.** [https://legis.senado.leg.br/dadosabertos/plenario/lista/votacao/\\$/langle\\$YYYYMMDD\\$/rangle\\$/langle\\$YYYYMMDD\\$/rangle\\$](https://legis.senado.leg.br/dadosabertos/plenario/lista/votacao/$/langle$YYYYMMDD$/rangle$/langle$YYYYMMDD$/rangle$). 1,767 plenary votes, 142,267 individual senator votes (2011–2025). Used to estimate coalition cohesion c_p (Section 2.2.5).
- **Câmara dos Deputados — Dados Abertos.** <https://dadosabertos.camara.leg.br/api/v2/votacoes>. 359 plenary votes from the 56th and 57th legislatures (2019–2026).
- **Câmara dos Deputados — Relatórios da Atividade Legislativa.** <https://www2.camara.leg.br/atividade-legislativa/plenario/relatorios-da-atividade-legislativa/dados-de-votacoes-nominais>. Reference URL provided by the project’s specification document.

1.13.6 Demographics, economics

- **IBGE — Sistema SIDRA.** <https://apisidra.ibge.gov.br/>. PNAD-C and Census 2022 tables (escolaridade 6407, rendimento 7434, desocupação 6318, população 4709, PIB per capita 5938, urbanização 1378).
- **IBGE — Pesquisa Nacional por Amostra de Domicílios (PNAD-C).** Reference page: <https://www.ibge.gov.br/estatisticas/sociais/populacao/9127-pesquisa-nacional-por-amostra-de-domicilios.html>.
- **IBGE — Censo Demográfico 2010, 2022.** Tables 137 and 200 (population by federation unit), table 1378 (urbanization rates).

1.13.7 Prediction markets

- **Polymarket — Gamma API.** <https://gamma-api.polymarket.com/markets>. Historical pricing for the contract “Lula wins 2022 Brazil election” and emerging 2026 contracts.
- **Kalshi — Trade API.** <https://api.elections.kalshi.com/trade-api/v2>. No liquid 2022 contracts; partial 2026 coverage.

1.13.8 State-level 2026 polls (aggregator)

- Maynard, L. (Paragon Capital). *Brasil Macro — Cenários Presidenciais 2026*. <https://rpubs.com/lucasmaynard/1418609>. Aggregator of state-level second-round polls for the Flávio Bolsonaro × Lula scenario (12 federation units, $\approx 79\%$ of population, April–May 2026). Includes the qualitative observation of Veritá’s right-leaning bias in SC 2022 (Lula 25.6 % poll vs 30.7 % real, -5.1 pp residual) used as a sanity check on §4 estimates.

1.13.9 Project-internal artifacts

All processing scripts and intermediate CSVs are version-controlled under `metodologia_eleitoral_br/` (see Appendix A for the directory listing). Each ingestion script documents its endpoint and the date of retrieval (May 20–21, 2026).

1.14 Software and Toolchain

- **Python 3.13** with the scientific stack: `numpy 2.4.6`, `scipy 1.17.1`, `pandas 3.0.3`, `scikit-learn 1.8.0`, `matplotlib 3.10.9`.
- **BeautifulSoup4** and **requests** for HTML scraping of poll aggregator pages.
- **Pandoc 3.9** + **Tectonic 0.16** for the LaTeX $\rightarrow$ PDF compilation pipeline of this paper.
- **React 18**, **Chart.js 4.4**, and **TailwindCSS** (via CDN) for the interactive companion dashboard (`cenarios_2T.html`).
- **Anthropic Claude (Claude 4.x family)** for drafting assistance in accordance with the disclosure preceding Section 1.

All numerical results in this paper are reproducible from the code and data in the project repository (Appendix A). For each input data file we record provenance (source URL, retrieval date, byte count) in the corresponding `README.md` of `data/`.

1.15 Appendix A — Reproducibility

All code, data, and results are organized in [metodologia_eleitoral_br/](#):

```
metodologia_eleitoral_br/
|-- src/
|   |-- model_2T.py          # closed-form posterior
|   |-- data_2T.py          # historical adapter
|   |-- data_2026.py        # 2026 succession scenarios
|   |-- house_effects.py    # partial pooling
|   `-- diagnostics_v4.py   # validation suite
|-- scripts/
|   |-- parse_polls_2T.py    # custom poll parser
|   |-- fetch_*.py          # data ingestion
|   `-- project_2026.py     # 2026 application
|-- data/
|   |-- tse/, polls/, ibge/  # raw sources
|   |-- approval/, senado/   # auxiliary
|   `-- local_rpubs/        # Maynard 2026 state-level polls
|-- results/
|   |-- backtest_2T.csv      # historical metrics
|   |-- projection_2026.json # 2026 marginal forecast
|   `-- diagnostics_v4/     # full diagnostic battery
`-- cenarios_2T.html        # interactive visualization
```

To reproduce all results:

```
python scripts/parse_polls_2T.py      # build polls_2T_clean.csv
python -m src.house_effects            # estimate biases
python backtest_2T.py                  # walk-forward backtest
python -m src.diagnostics_v4           # full diagnostic battery
python -m scripts.project_2026        # 2026 forecast
python build_html_2T.py                # render dashboard
```

Working paper. Comments welcome.